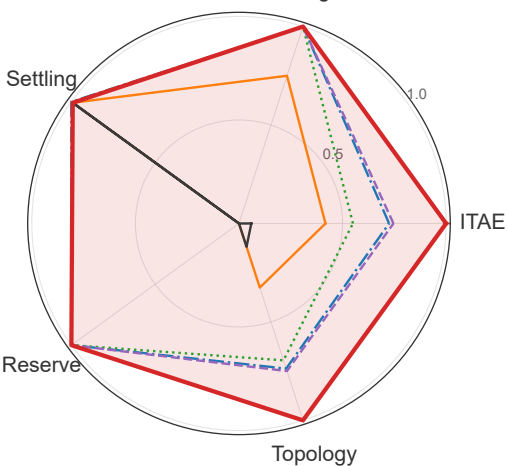
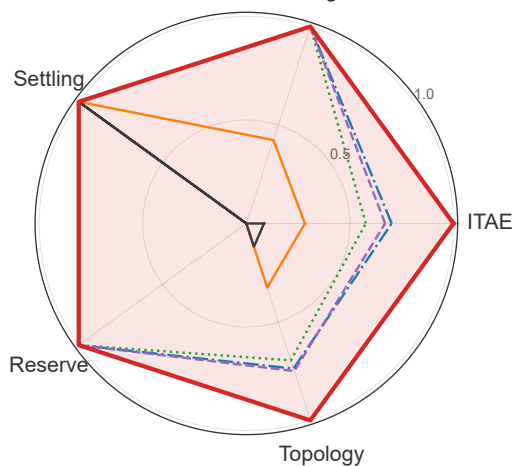


— GraphSAGE (ours) - - - MATD3 ···· GCNN-PPO - - - MLP-MAPPO — Fixed droop — No FFR

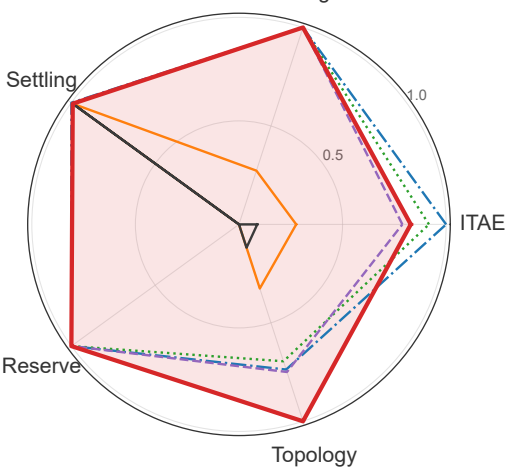
S1 load step (+2.5 MW)
Margin



S2 gen trip (-3.9 MW)
Margin



S3 line trip (-2.4 MW)
Margin



S4 high-ren surge (+4.7 MW)
Margin

